FYUGP B.A. (HONS) IN ECONOMICS / B.A. (HONS.) IN ECONOMICS

(RESEARCH) UNDER THE HONORS SCHOOL SYSTEM FOR THE SESSION 2023-24

1. **Outlineof Syllabi and Course Structure for FYUGP B.A.(Hons) in Economics / B.A.(Hons.) in Economics (Research) under the Honors School System for the Session 2023-24 in the Departmentof Economics**
2. **B.A. First year Template as per NEP**

NEP Outlines of Syllabi and Course Structure for various Courses in the Department ofEconomics 2023-24.

B.A. (Hons.) Semester I (under NEP)

Economics Core Course 1: Introductory Microeconomics(4 credits)

Economics Core Course 2: Mathematical Methods forEconomics-I (4 credits)

Skill Enhancement Course I:Data Collection and Handling Techniques (3 credits)/**Any other course offered by Maths / Statistics / Computer Science Departments of the Panjab University of same or more credits**

Multidisciplinary Course 1: Fundamentals of Sociology (3 credits)/**any other course offered by the Panjab University of same or more credits**

Ability Enhancement Compulsory Course (AEC) –I (2 credits) (offered by English Department)

Value Added Course 1: Indian Economic Thought (2 credits)/**any other course offered by the Panjab University of same or more credits**

Value Added Course 2: Introduction to data Analysis Using Excel-1 (2 credits)/**Any other course offered by Maths / Statistics / Computer Science Departments the Panjab University of same or more credits**

Minor 1 (any one)

STATISTICAL METHODS (4 credits)/**Any other course offered by theStatistics Department of same or more credits**

Or

Calculus(4 credits)/**any other course offered by Maths. Department of same or more credits**

B.A. (Hons.) Semester II (under NEP)

Economics Core Course 3: Introductory Macroeconomics –II(4 credits)

Economics Core Course 4: Mathematical Methods forEconomics-II(4 credits)

Skill Enhancement Course 2:**Data Analysis for Beginners (3 credits)**

/ **Any other course offered by Maths / Statistics / Computer Science Departments the Panjab University of same or more credits**

Multidisciplinary Course 2: Social Stratification (3 credits)/**Any other course offered by the Panjab University of same or more credits**

Ability Enhancement Compulsory Course (AECC) –II (2 credits) (offered by English Deptt.)

Value Added Course 3: Introduction to Environmental Economics
(2 credits)/**Any other course offered by the Panjab University of same or more credits**

Value Added Course 4:Introduction to data Analysis Using Excel-II (2 credits)/ **Any other course offered by Maths / Statistics / Computer Science Departments the Panjab University of same or more credits**

Minor 2 (any one)
Introductory Probability (4credits)/**Any other course offered by Statistics Departments the Panjab University of same or more credits**

Or

Algebra (2 credits) + **Differential Equations (2credits)/Any other course offered by the Maths Department, Panjab University of same or more credits**

Course Structure and Syllabi for FYUGPB.A. (Hons.) Economics under the Honours School System for 1st and 2nd Semester for 2023-24 Session

Course Structure for FYUGPB.A. (Hons.) Economics under the Honours School System for 1st and 2nd Semester for 2023-24 Session

Semester-I			
Course	**Course Code**	**Paper**	**Credits**
Economics Core Course 1	BAHSECO-101	Introductory Microeconomics	4
Economics Core Course 2	BAHSECO-102	Mathematical Methods for Economics-I	4
Skill Enhancement Course 1	BAHSECOSEC-101	Data Collection and Handling Techniques	3
Multidisciplinary Course 1	MDSCSOC-101	Fundamentals of Sociology	3
Ability Enhancement Compulsory Course 1		English	2

Value Added Course 1	**BAHSECOVAC-101	Indian Economic Thought	2
Value Added Course 2	BAHSECOVAC-102	Introduction to data Analysis Using Excel-I	2
Minor (any one):			
Statistics Course 1	*STAT-M-I11(Theory) + STAT-M-I12(Practical)	Statistical methods	4
Mathematics Course 1:	*MAT-MI-111	Calculus	4
Total Credits			24
Semester-II			
Economics Core Course 3	BAHSECO-103	Introductory Macroeconomics	4
Economics Core Course 4	BAHSECO-104	Mathematical Methods for Economics-II	4
Skill Enhancement Course 2	BAHSECOSEC-102	Data Analysis for Beginners	3
Multidisciplinary Course 2	MDSCSOC-102	Social Stratification	3
Ability Enhancement Compulsory Course 2		course offered by English Dept.	2
Value Added Course 3	**BAHSECOVAC-103	Introduction to Environmental Economics	2
Value Added Course 4	BAHSECOVAC-104	Introduction to data Analysis Using Excel-II	2
Minor (any one)			
Statistics Course 2	*STAT-M-151(Theory) + STAT-M-152 (Practical)	Introductory Probability	4
Mathematics Course 2	*MAT-MI-161 + *MAT-MI-162	Algebra + Differential Equations	4
Total Credits			24

****BAHSECO-101(4 credits) +BAHSECO VAC-101(2 credits) would be offered as Minor-1 (6 credits) to Non Economics Honors students in the Semester-I**

****BAHSECO-103 (4 credits) + BAHSECOVAC-102 (2 credits) would be offered as Minor-2(6 credits) to Non Economics Honors students in the semester-II**

Note:

*** 1. A 6 credit minor (150 marks) would be treated at par with a 4 credit minor (100 marks)and 150 marks to be normalized to 100 for BA. Thus Maths and Stats minor students would have more than 4 teaching hours every week.**

**** 2. Regarding balancing of BSc BA Minor credits the first year**

A B.Sc. student who opts for a minor of 4 credits from B.A. will cover the remaining 2 Credits from the VAC of the same subject of 2 credits thus earning total 6 credits in that minor and would have two VAC courses in the first and second semester. In the first semester 1 to 2 VAC courses are allowed.

Syllabi of B.A. (Hons.) Semester- I (under NEP)

BAHSECO-101: INTRODUCTORY MICROECONOMICS

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective:-This course is designed to expose the students to the basic principles of microeconomic theory. The emphasis will be on thinking like an economist and the course will illustrate how microeconomic concepts can be applied to analyze real-life situations.

Course Outcomes: -1. To train them to think like an economist. 2.To rigorously and comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms and markets using, ingeneral, a static and partial equilibrium framework.

Pedagogy of the Course Outcomes:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:-

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60)

UNIT- I

Exploring the subject matter of Economics. Why study economics? Scope and method of economics; the economic problem: scarcity and choice; the question of what to produce, how to produce and how to distribute output
Cardinal Utility Analysis.
Ordinal Utility Analysis: The consumption decision - budget constraint, consumption and income/price changes, demand for all other goods and price changes; description of preferences (representing preferences with indifference curves); properties of indifference curves; consumers optimum choice; income and substitution effects.

UNIT- II

Theory of production: laws governing short run and long run. Producers equilibrium and expansion path
Cost Theory: Short run and long run cost curves.
The Firm and Industry under Perfect Market Structure: Price and output determination; short run and longrun.

UNIT- III

Determinants of individual demand/supply; demand/supply schedule and demand/supply curve; market versus individual demand/supply; shifts in the demand/supply curve, demand and supply together; how prices allocate resources. The basic competitive model. Supply and Demand How Markets Work, Markets and Welfare Markets and competition Elasticity of demand: its measurements, applications and determinants.Consumer surplus; producer surplus and the efficiency of the markets.
Concept of dead weight loss; Govt. intervention in the form of Taxes and Subsidies.

UNIT- IV

Single seller monopoly: Price and output determination
Input Markets Labour and land markets - basic concepts (derived demand, productivity of an input, marginal productivity of labour, marginal revenue product); demand for labour; input demand curves; shifts in input demand curves; competitive labour markets and firms profit maximisation conditions in input market ; and labour markets and public policy. Derivation of labour

supply and savings decision - choice between leisure and consumption.

Essential Readings:-

1. N. Gregory Mankiw. (2007) *Economics: Principles and Applications*. (4th ed.). India edition by South Western, a part of Cengage Learning, Cengage Learning India PrivateLimited.
2. Karl, E. Case., and Ray C. Fair, (2007). *Principles of Economics (*8th ed.). Pearson Education Inc.

Suggested Readings:-

1. Joseph E. Stiglitz, and Carl E. Walsh. (2007) .*Economics* (4th ed.). W.W. Norton & Company, Inc., New York, International StudentEdition.
2. Salvotore. D (2006) *Theory and Problems of Microeconomic Theory* (Sachems series) (3rd ed.) Tata McGraw-Hill Publishing CompanyLtd.
3. Salvotore.D. (2007). *Micro economic theory and Applications* (4th ed.) Oxford University Press.

BAHSECO-102: MATHEMATICAL METHODS FOR ECONOMICS–I

Topology of the code	**Major**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective:-This is the first of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in this syllabus. In this course, particular economic models are not the ends, but the means

for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication at which the material is to be taught is indicated by the contents of the prescribed textbook.

Course Outcomes : - 1. Students are able to understand the preliminaries and functions of one real variable and their economic applications. 2. Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3. Students are able to understand and apply the integration o functions and their simple economic applications. 4. Students are able to understand and apply the Difference Equations and their simple economic applications.

Pedagogy of the Course Outcomes:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT- I

Preliminaries and Functions of one real variable: Logic and proof techniques; sets and set operations; relations; functions and their properties; number systems. Graphs; elementary types of functions: quadratic, polynomial, power, exponential, logarithmic; sequences and series: convergence, algebraic properties and applications; continuous functions: characterizations, properties with respect to various operations and applications.

UNIT- II

Differentiable Functions and Single-variable optimization: Differentiable functions: characterizations, properties with respect to various operations and applications; second and higher order derivatives: properties and applications. Geometric properties of functions: convex functions, their characterizations and applications; local and global optima: geometric characterizations, characterizations using calculus and applications.

UNIT- III

Integration of functions: Methods of Substitution and partial fractions and simple economic applications

UNIT- IV

Difference equations: Introduction, solution of difference equations upto 2nd order, simple economic applications.

Essential Readings:-

K. Sydsaeter and P. Hammond, (2002).*Mathematics for Economic Analysis*, Pearson Educational Asia: Delhi.

Suggested Readings:-

Weber, Jean E (1982). Mathematical *Analysis Business and Economic Applications* (4^{th}Ed.)Yamane

Taro(1972). Mathematics for Economists.

BAHSECO SEC-101: Data Collection and Handling Techniques

Topology of the code	**Skill Enhancement Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment 15 marks**
Time	**3 hours**

Course Objective:- To provide skills to undertake data based research so that students get the competency of executing sample surveys and knowledge about secondary data collection.

Course Outcomes : -It will enable the students to qualified for jobs relating to field investigation in any field of social sciences.

Pedagogy of the Course Work:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question paper in this regard.

- There shall be**7**questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of 3marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT-I

Research: Meaning, objectives, Characteristics, Types, Process and Utility, Research method Vs. Methodology. Research Design-Meaning, Types and Features of Good Research Design. Types of Data: Primary vs Secondary**,** Cross Section or Micro Data, Time Series Data, Pooling of data, Panel data, Censusvs sampled data.

UNIT- II

Sampling – Concepts of Statistical Population, Sample, Sampling Frame, Sampling and non-Sampling Errors, Sample Size. Census and Sampling survey.Characteristics of good sampling.
Sampling Techniques: Simple Random Sampling; PPS Sampling; Stratified Random Sampling; Multistage Sampling (Applications only), determination of sample size.

UNIT-III

Methods of collection of Primary data (observation method, interview method, collection of data through questionnaires- Designing of questionnaire, collection of data through schedules- Designing of schedules).Measurement and scaling, validity and reliability.
Collection of Secondary Data
Selection of appropriate method of data collection.

**** Practical Work: Students are required to formulate a research problem and need to prepare a questionnaire for the same along with appropriate measurement and scaling techniques.

Essential Readings:-

1. Kothari, C.R. (2009). *Research Methods and Techniques*, Wiley Eastern.

Suggested Readings:-

1. Bethlehem, J. (2009). *Applied survey methods: A statistical perspective.* Wiley.
2. Cochran, W. (2008). *Sampling techniques, 3rd ed.* Wiley.
3. Cooper, D., Schindler, P., Sharma, J. (2012). *Business research methods*, *12th ed.* McGraw-Hill.
4. Flick, U. (2012). *Introducing research methodology: A beginner's guide to doing a research project*. Sage Publications.
5. Groves, R., Fowler, F., Couper, M., Lepkowski, J., Singer, E., Tourangeau, R. (2009). *Survey Methodology*.Wiley.
6. Kumar, R. (2014). *Research methodology: A step by step guide for beginners*, *4th ed.* Sage Publications.

MDSCSOC-101: FUNDAMENTALS OF SOCIOLOGY

Topology of the code	**Multidisciplinary Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment 15 marks**
Time	**3 hours**

Course Objective: - The systematic study of human behaviour and human society is a relatively recent development. The central emphasis of the paper would be to develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to a development of a sociological terms and themes, thus leading to a development of a sociological outwork. The course would also discuss the works of founding father which led to the development to sociological thinking.

Course Outcomes : -1. Students are able to study of human behaviour and human society relative to recent development. 2. Students develop a conceptual clarity regarding the basic sociological terms and themes, thus leading toa development of sociological terms and themes, thus leading to a development of a sociological outwork.

Pedagogy of the Course Work:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question paper in this regard.

- There shall be**7**questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of 3marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT – I

Developing Sociological Outlook*:* Definition and Scope; Significance and Uses of Sociology; Relationship of Sociology with other Social Sciences – Anthropology, Economics, Political Science, Psychology, Philosophy and History.

UNIT – II

Basic Concepts: **Society** – Meaning and Characteristics, Theories of Origin, Individual and Society. Community, Association and Institution – Meaning, Characteristics and Differences.Groups – Definition, Characteristics and Classification.

UNIT – III

Culture and Socialization: Culture-Definition, Cultural Lag, Acculturation, Assimilation, Pluralism. Culture and Personality.
Civilization: Definition, Traditional and Modern. Socialization – Meaning, Stages, Agencies.

Essential Readings:-

1. Giddens, Anthony .*Sociology, Cambridge (*4th ed.). Polity Press.

Suggested Readings:-

2. Haralambos, Michael.(2015). *Sociology: Themes and Perspectives*.Oxford.
3. Davis, Kingley.(2002). *Human Society.* Surjeet Publications,Delhi.
4. Maclver, R S & Page Charles. *Society: An Introductory Analysis* (Latest ed.). Macmillan India Ltd.
5. Johnson, Harry (2003). *Sociology: A Systematic Introduction*. Allied Publishers, NewDelhi.

6. Inkeles Alex. (2004).*What is Sociology? An Introduction to the Discipline and Profession*. Prentice-Hall,India.
7. Bradshaw, York W, et.al. (2001). *Sociology for a New Century*, Piner Forge Press, California.
8. Johnson, Doley Paul.(1986).*Sociological Theory* (Or later ed.). New York,McMillan.
9. Ritzer, George. (2011).*Sociological Theory* .London (Or latest ed.). McGraw-Hill.
10. CoserLewis. (2004). *Masters of Sociological Thought*, Rawat Publications, NewDelhi.
11. Rao Shankar C.N. (2012).*Sociology*: *principles with an Introduction to Social Thought*.S.Chand and Company Pvt.Ltd.

BAHSECOVAC-101:INDIAN ECONOMIC THOUGHT

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours(2 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Course Objective:- The course aims at enabling the students to acquire and demonstrate the knowledge on economic thought with its historical perspective.

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Course Outcomes : - 1. Students through this course become familiar with the historical evolution of economic ideas.The area able to learn the foundational contributions of the classical economists to the economic literature. 2. Studentsareabletodevelopunderstandingaboutthephilosophicalfoundationsofeconomic analysis of capitalist system. They also learn the theories of imperialism which areimportanttounderstandtheexploitationprocessofthirdworldcountriesbythemetropolisnation s.3. Students are able to the origin of marginalist, neoclassical and Austrian school of thought and their main features. This is important to understand the differences and similarities among their ideas. 4. Students are able to learn about the Keynesian revolution in economic theory. They also understand the causes of great depression. Moreover, they learn about the Kaleckian version of macro economics.

Pedagogy of the Course Work:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:
• The maximum marks for the paper will be 50. The question paper will be of 40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.
• There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from eachunit. Each question will carry 10 marks (15x2=30).

UNIT-I

Pre-independence Indian Economic thought
Kautilya'sArthasastra: The economic functions of the state, Taxation and pricing policy, The land system

DadabhaiNaoroji and the drain theory: What is the drain?, Criticisms and Long-run relevance

Ranade and the economics of development: Poverty and industrialisation, Agrarian policy, Railway investment.

UNIT-II

Indian Economic thought towards independence
Gokhale and the economics of education

Gandhian economics: Methodological issues, Consumption behaviour: the limitation of wants, Swadeshi; Trusteeship and industrial relations; Charity, leisure and the sanctity of work

Essential Readings:-

1. Dasgupta, A.K. (1993). A History of Indian Economic Thought (1st ed.). Routledge.

Suggested Readings:-

2. Kautilya. &Shamasastry, R. (1967). *Kautilya'sArthasastra*. Mysore, India : Mysore Publishing and Printing House.
3. Naoroji, D. (1901). Poverty and Un-British Rule in India. United Kingdom: S. Sonnenschein.
4. Chandra, B., Ranade, M. G. (1990). Ranade's Economic Writings. India: Gian Publishing House.
5. Gokhale, G. K. (2018). Gopal Krishna Gokhale: Select Speeches and Writings. India: National Book Trust, India.
6. Village Swaraj by M.K. Gandhi. (2021). (n.p.): PrabhatPrakashan.
7. Dasgupta, A. K. (1996). Gandhi's Economic Thought. (n.p.): Taylor & Francis.

BAHSECOVAC-102 : INTRODUCTION TO DATA ANALYSIS USING EXCEL-I

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours(2 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoning involved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjective type questions/ answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Instructions for Paper-setter and candidates:

• The maximum marks for the paper will be 50. The question paper will be of 40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.

• There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 10 marks (15x2=30).

Course Outcomes

After successful completion, the students will be able to understand the use of MS-Excel in data handling and presenting. The filtering techniques, lookup methods, pivot tabling, graphics presentation, matrix analysis etc. shall sharpen a student in the field of quantitative economics and econometrics.

Unit-I

Introduction to Spreadsheets

Introduction to spreadsheets, reading data, manipulating data.Basic spreadsheet operations and functions.

Spreadsheet Functions to Organize Data

Introduction to some more useful functions such as the IF, nested IF, VLOOKUP and HLOOKUP functions in Excel.

Unit-II

Introduction to Filtering, Pivot Tables, and Charts

Introduction to the Data filtering capabilities of Excel, the construction of Pivot Tables to organize data and introduction to charts in Excel.Constructing various Line, Bar and Pie charts.Using the Pivot chart features of Excel.Understanding and constructing Histograms and Scatterplots.

References:

Lloyd, Chris, J. (2011). Data Driven Business Decisions, John Wiley & Sons.

Levine, D.M., Stephen, D. and Szabat, K.A. (2016).Statistics for Managers Using Microsoft Excel, Pearson Education.

Day, Alastair L. (2013). Mastering Financial Mathematics in Microsoft Excel, Prentice Hall.
Piascik, Chester (2001). Applied Mathematics with Microsoft Excel, Pennsylvania State University.

Jelen, Bill (2013). Excel 2013: Charts and Grap Microsoft Excel. Que Publishing.

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STAT-M-111: STATISTICALMETHODS

Programme Outcome: The course aims to shape the attitudes of learners regarding the field of statistics**.** The objective of the course is to make the students familiar with various techniques used in summarization, presentation and analysis of data. The focus will be both on theoretical as well as practical approach.						
Course Outcome: On completion of this course students will be able to: (i) Get familiar with various techniques used in summarization and analysis ofdata. (ii) Understand the nature of statisticaldata. (iii) Understand about the types of correlations for bivariate and multivariate dataand their uses in real lifeproblems. (iv) Highlight the concept of linear regression and its applications in reallife (v) Understand the various measures of association betweenattributes.						
Class	**Sem.**	**CourseCode**	**Course Title**	**Credits= 6**		
B.A./B.Sc.	**I**	**STAT-M-111**	**STATISTICALMETHODS**	**L=4**	**T=1**	**P=1**

Total Marks (Theory= 100) (Tutorial +Practical= 50) Internal Assessment= 20%	
Total Hours= 105 hours (Theory= 60 hours +Tutorial=15 hours +Practical= 30 hours)	
Time Allowed for Examination: Theory= 3 hours; Practical= 3 hours	
Instructions for Paper setter	There will be in all nine (9) questions, all of equal marks. The first question is **compulsory** and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit
Instructions for Candidate	The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each Unit**.**

Course Contents:

UNIT I

Introduction: Definition and scope of Statistics, concepts of statistical population and sample. Scales of measurement - nominal, ordinal, interval and ratio. Variables and attributes, Diagrammatical Representation of Data, Summarization of Data: Frequency Distribution and Graphical Presentation (histogram, frequency polygon and ogives).

Measures of Central Tendency: mathematical and positional. Measures of Dispersion: range, quartile deviation, mean deviation, standard deviation, coefficient of variation, moments, measures of skewness and kurtosis and Box-whisker Plot.

UNIT II

Bivariate data: Definition, scatter diagram, simple correlation, rank correlation (with ties and without ties). Partial and Multiple Correlation Coefficients. Only definition and its applications (No derivation)

Fitting of linear and quadratic regression equations using principle of least squares.Theory of attributes and consistency of data, independence and association of attributes, measures of association and contingency for 2 x 2 and r x s contingency tables.

Notes:

1. **This course is not for the students of B.A./B.Sc. (General) Statistics Discipline Specific courses(Major/Core).AnyothercandidateshallofferthisminorsubjectinB.A./B.Sc.**
2. **Students are to be engaged to 7 hours of teaching per week (4 lectures for theory, one lecture for tutorial and 2 lectures forpractical).**
3. **Practical Examination will be held in the respective/samesemester.**

SUGGESTEDREADINGS:

1. Goon,A.M.,Gupta,M.K.andDasgupta,B.(2013):FundamentalofStatistics,VolI,WorldPress, Kolkata
2. Goon,A.M.,Gupta,M.K.andDasgupta,B.(2011):FundamentalofStatistics,VolII,WorldPress, Kolkata.
3. Miller, Irwin and Miller, Marylees (2006): JohnE. Freund's Mathematical Statistics with Applications, (7th Edn.), Pearson Education,Asia.
4. Mood, A.M. Graybill, F.A. and Boes, D.C. (2011): Introduction to the Theory of Statistics, 3rd Edn., (Indian Edition), Tata McGraw-Hill Pub. Co.Ltd.

STAT-M-112(PRACTICAL): STATISTICAL METHODS

(Credit: 1)

Practical: Max. Marks: 25

Practical will be based on Paper STAT-M-112 (PRACTICAL): STATISTICAL METHODS

Practical: Final Written Examination: 15 marks, Internal Assessment: 5 marks, viva-voce: 5 marks

Note for Practical: The practical question paper will consist of 5 questions. The students will be required to attempt any three questions in three hours duration.

MAT-MI-111: Calculus

Credits: 6 (L=5; T=1; P=0) Total hours: 90 (Theory=75 + Tutorials =15)
Total Marks: 150 (Including Internal Assessment= 30) Time Allowed for Examination: 3hrs.

Instructions for the Candidates and Paper Setters

Candidates will be asked to attempt five questions out of nine, carrying equal marks. Question No.1 spread
over the whole syllabus will be compulsory.
There will be two questions from each unit and the students will have to attempt one from each unit.

Course Outcome: The main goal of this course is to deliver the basics of sequences and series; differential and integral calculus, for real as well as multivariate functions. It is expected that the students develop a taste of writing proofs, particularly for Unit I and Unit III, rather than applying formulas only.

Unit-I

Differential Calculus: Precise definition of limit, continuity, one-sided limit, limits involving infinity, asymptotes of graphs, tangents and the derivative at a point, the derivative of a function, extreme values of functions, mean value theorem, monotone functions and the first derivative test, test for concavity, tracing of curves.
(Scope: Sections 2.3 { 2.6, 3.1, 3.2, 4.1 { 4.4 of (A)).

Unit-II

Integral Calculus: Riemann sums, definite integrals, area between curves, volumes using cross sections and cylindrical shells, arc length and areas of surfaces of revolution.
(Scope: Sections 5.1, 5.6, 6.1 {6.4 of (A)).

Unit-III

Infinite sequences and Series: sequences, in nite series, the integral test, comparison test, the root and ratio test, alternating series, absolute and conditional convergence, power series, Taylor and Maclaurin series, Power series, convergence of Taylor series, application of Taylor series.
(Scope: Sections 10.1 - 10.10 of (A).)

Unit-IV

Polar Coordinates: Polar Coordinates, graphing in Polar coordinates, areas and lengths in Polar coordinates. Multivariable Functions: Limits and continuity for functions of several variables, partial derivatives, the chain rule, directional derivatives, gradient vectors, tangent planes, extreme values and saddle points. (Scope: Sections 11.3-11.5, 14.2 { 14.7 of (A)). 9

Essential Textbooks

(A) G. B. Thomas, M. D. Weir and J. R. Hass, Thomas' Calculus, (12th ed.), Pearson Education, 2014.

Further Readings

1. J. L. Taylor Foundations of Analysis, American Mathematical Society, 2012.
2. S. Narayan, Integral Calculus, S. Chand and Company Ltd, 2001.
3. M.J. Strauss, G.L. Bradley and K. J. Smith, Calculus, (3rd ed.), Pearson Education, 2007.
4. H. Anton, I. Bivens and S. Davis, Calculus, (7th ed.), John Wiley and Sons, 2002.
5. R. Courant and F. John, Introduction to Calculus and Analysis (Volumes I & II), Springer-Verlag, Inc., 1989.

SEMESTER- II

BAHSECO-103: INTRODUCTORY MACROECONOMICS

Topology of the code	**Major/Minor**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective:-This is the first module in a three-module sequence that introduces students to the basic concepts of Macroeconomics. Macroeconomics deals with the aggregate economy. This course discusses the preliminary concepts associated with the determination and measurement of aggregate macroeconomic variables like GDP, savings, investment, money, inflation and the balance of payments. It also introduces students to simple analytical frameworks (e.g., the IS-LM model) for determination of equilibrium output.

Course Outcomes: - This is the first modules in a three modules sequence that introduces students

to the basic concepts of macroeconomics. 1. Students are able to understand to basic concepts and fundamentals of macro economics necessary for a proper understanding of the discipline. 2.Students are able to understand and preliminary concepts associated withth edetermination and measurement of aggregate macroeconomics variables like savings, investment, GDP, money, inflation, balance of payments and simple in come determination of a closed economy. 3. Students are able to understand simple analytical frameworks for determination of equilibrium output in the closed economy in the short run as well as in the long run and the role of policy in this contest.

Pedagogy of the Course Work:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions in all.

The first question would be compulsory, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidates will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10 x 2=20).

Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidate shall be required to attempt one question from each unit-4 in all. Each question shall carry 15 marks (15 x 4= 60)

UNIT-I

Introduction to macroeconomics and national income accounting Basic issues studied in macroeconomics: Measurement of gross domestic product; income, expenditure and the circular flow; real versus nominal GDP; price indices; national income accounting for an open economy; balance of payments: current and capital accounts.

UNIT-II

Money Functions of money; quantity theory of money; determination of money supply and demand; credit creation; tools of monetary policy.

UNIT-III

Inflation Types; Social Costs;

Hyperinflation.

UNIT-IV

The closed economy in the shortrun

Classical and Keynesian systems; simple Keynesian model of income determination; IS-LM model; fiscal and monetary multipliers

Essential Readings:-

1. Abel, A., Bernanke, B. (2016). Macroeconomics, 9th ed. PearsonEducation.
2. Blanchard, O. (2018). Macroeconomics, 7th ed. PearsonEducation.

Suggested Readings:-

3. Dornbusch, R., Fischer, S., Startz, R. (2018). Macroeconomics, 12th ed.McGraw-Hill.
4. Jones, C. (2016). Macroeconomics, 4th ed. W. W.Norton.
5. Mankiw, N. (2016). Macroeconomics, 9th ed. WorthPublishers.

BASHECO-104: MATHEMATICAL METHODS FOR ECONOMICS –II

Topology of the code	**Major**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 60 hours(4 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**4**
Max. Marks	**100 Marks = Theory 80 marks + Internal Assessment 20 marks**
Time	**3 hours**

Course Objective :-This is the first of a compulsory two-course sequence. The objective of this sequence is to transmit the body of basic mathematics that enables the study of economic theory at the undergraduate level, specifically the courses on microeconomic theory, macroeconomic theory, statistics and econometrics set out in this syllabus. In this course, particular economic models are not the ends, but the means for illustrating the method of applying mathematical techniques to economic theory in general. The level of sophistication at which the material is to be taught is indicated by the contents of the prescribed textbook.

Course Outcomes : - 1. Students are able to understand the preliminaries and functions of one real variable and their economic applications. 2. Students are able to learn about differentiable functions and also single variable optimization using calculus and its applications. 3. Students are able to understand and apply the integration o functions and their

simple economic applications. 4. Students are able to understand and apply the Difference Equations and their simple economic applications.

Pedagogy of the Course Outcomes: - 80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 100. The question paper will be of 80 marks and continuous evaluation 20 marks. Time allowed will be 3hours.

The paper-setter must put a note in the question paper in this regard.

- There shall be **9** questions inall.

The first question **compulsory** comprising 15 short answer type questions spread over the whole syllabus. The candidates are required to attempt 10 questions. Each question shall be of **two** marks (10 x 2= 20).

Rest of the paper shall contain four units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry 15 marks (15x4=60).

UNIT- I

Differential Equation*:* Introduction; Solution: Variable Separable Case, Homogeneous Case, Standard Linear Differential Equation, Bernoulli's form, Exact Equation; Solution of Linear differential Equation with Constant Coefficients; Simple Application Questions.

UNIT-II

Linear algebra: Vector spaces: algebraic and geometric properties, scalar products, norms, orthogonality; linear transformations: properties, matrix representations and elementary operations; systems of linear equations: properties of their solution sets; determinants: characterization, properties and economicapplications.

UNIT- III

Functions of several real variables: Geometric representations: graphs and level curves; differentiable functions: characterizations, properties with respect to various operations and applications; second order derivatives: properties and applications; the implicit function theorem, and application to comparative statics problems; homogeneous and homothetic functions: characterizations and economic applications.

UNIT- IV

Multi-variable optimization: Convex sets; geometric properties of functions: convex functions, their characterizations, properties and applications; further geometric properties of functions: quasi convex functions, their characterizations, properties and applications; unconstrained optimization: geometric characterizations, characterizations using calculus and applications; constrained optimization with

equality constraints: geometric characterizations, lagrange characterization using calculus and applications; properties of value function: envelope theorem and economic applications.

Essential Readings:-

K. Sydsaeter and P. Hammond (2002). *Mathematics for Economic Analysis*, Pearson Educational Asia Delhi.

Suggested Readings:-

Weber,JeanE(1982*).* *MathematicalAnalysisBusinessandEconomicApplications*(4th Ed.)Yamane, Taro(1972). *MathematicsforEconomists*.

BASHECOSEC -102: DATA ANALYSIS FOR BEGINNERS

Topology of the code	**Skill Enhancement Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment 15 marks**
Time	**3 hours**

Course Objective:-To provide skills to undertake data based research so that students get the competency of executing sample surveys and knowledge about secondary data collection.

Course Outcomes: -It will enable the students to qualified for jobs relating to field investigation in any field of social sciences.

Pedagogy of the Course Work:- 80% Class room teaching.

20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question paper in this regard.

- There shall be**7**questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of **3** marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT-I

Collection of primary data through questionnaire (online (google form)/ face to face)
Errors in surveys: Misunderstanding of question and answers, problem of non-response.
Processing of Data: Questionnaire checking, Editing, Coding, classification, Tabulation, Graphical representation (Bar charts, Pie Charts, line graphs) Data cleaning, Data Adjusting.

UNIT- II

Analysis of Data and Interpretation: Measures of central tendency (Arithmetic mean median and mode), Dispersion, simple correlation (Karl Pearson and Spearman Rank Correlation) and simple regression.

UNIT- III

Preparation of a Report: Types of Report, Research Report Format, Principles of Writing Reports, Documentation, Footnotes and References & Bibliography.
Use of computers in research.
Ethics and scientific integrity: standards of conduct, privacy in data, Plagiarism.

****Practical Work: Project Report (collection data through questionnaire on a research problem and preparation, analysis and interpretation of data) .

Essential Readings:-

1. Bethlehem, J. (2009). *Applied survey methods: A statistical perspective.* Wiley.
2. Cochran, W. (2008). *Sampling techniques, 3rd ed.* Wiley.

Suggested Readings:-

1. Cooper, D., Schindler, P., Sharma, J. (2012). *Business research methods*, *12th ed.* McGraw-Hill.
2. Flick, U. (2012). *Introducing research methodology: A beginner's guide to doing a research project*. Sage Publications.
3. Groves, R., Fowler, F., Couper, M., Lepkowski, J., Singer, E., Tourangeau, R. (2009). *Survey Methodology*.Wiley.
4. Kothari, C.R. (2009). *Research Methods and Techniques*, Wiley Eastern.

5. Kumar, R. (2014). *Research methodology: A step by step guide forbeginners*, *4th ed.* SagePublications.

MDSCSOC-102: SOCIAL STRATIFICATION

Topology of the code	**Multidisciplinary course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 45 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**3**
Max. Marks	**75 Marks = Theory 60 marks + Internal Assessment 15 marks**
Time	**3 hours**

Course Objective: -The systematic study of human behavior and human society is a relatively recent development. The central emphasis of the paper would be to develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to a development of sociological terms and themes, thus leading to a development of a sociological outwork. The course would also discuss the works of founding father which led to the development to sociological thinking.

Course Outcomes : - 1. Students are able to study of human behavior and human society relative to recent development. 2. Students develop a conceptual clarity regarding the basic sociological terms and themes, thus leading to a development of sociological terms and themes,thus leading to a development of a sociological outwork.

Pedagogy of the Course Work:-80% Class room teaching.
20% Internal Assessment.

Instructions for Paper-setter and candidates:

- The maximum marks for the paper will be 75. The question paper will be of 60 marks and internal assessment 15.

The paper-setter must put a note in the question paper in this regard.

- There shall be**7**questions in all.

The first question is **compulsory** comprising 8 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of **3** marks (5x3=15).

Rest of the paper shall contain three units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from each unit. Each question will carry15 marks. (15x3=45).

UNIT – I

Social Stratification: Concept - Social Stratification, Inequalities and Differentiation. Elements: Position, Role and Status.

UNIT II

Forms of Stratification and Social Mobility: Forms of Stratification – Caste, Class, Race and Gender. Social Mobility: Definition; Indicators – Social, Economic, Political; Types – Horizontal and Vertical (Upward and Downward) and Intergenerational.

UNIT – III

Social Change: Social Change – Definition. Factors: Demographic, Technological, Educational, Legislative. Processes of Change: Sanskritization, Westernization, Modernization and Globalization.

Essential Readings:-

1. The New Middle Class in India – A Sociological Analysis, Jaipur, Rawat.
2. Singh, Yogendra (1986). *Modernization of Indian Tradition*, Jaipur, Rawat (or latested.)
3. Srinivas, M N. (2010) .*Social Change in Modern India*, Orient Black Swan Pvt.Ltd.
4. Sharma, K L. (2006). *Social Stratification and Mobility*, Rawat Publication, New Delhi.
5. Abraham Francis, M. (2004). *Modern Sociological Theory – An Introduction*, Oxford University Press, Delhi.

Suggested Readings:-

1. D'Souza, V. S. (1981). *Inequality and Its Perpetuation*, Manohar Publications, New Delhi.
2. Singh, K S. (ed.) (1992). *Ethnicity Caste and People: India and Soviet Union.* Manohar Publications, New Delhi.
3. Singer Milton, CohaBeemard (2001). *Structure and Change in Indian Society*, Rawat Publications, New Delhi.
4. AtalYogesh.(2006). *Changing Indian Society*, Rawat Publications, NewDelhi.
5. Beteille, Andre (2009). Sociology – Essays on Approach & Method. Oxford University Press.
6. *Rao Shankar* C.N. (2012).*Sociology*: principles with an Introduction toSocial Thought.S.Chand and Company Pvt. Ltd.
7. *Rao Shankar* C.N. (2012). SOCIOLOGY OF INDIAN SOCIETY.S.Chand and Company Pvt.Ltd.
8. Haralambos, Michael.(2015). Sociology: Themes and Perspectives. Oxford.
9. Ministry of Women and Child Development, GOI. Report of the High Level Committee on Status of Women in India (2015).http://www.wcd.nic.in/documents/hlc-status-women
10. Johnson, DoleyPaul.(1986).Sociological Theory (Or later ed.). New York,McMillan.
11. Ritzer, George. (2011).Sociological Theory .London (Or latest ed.). McGraw-Hill.
12. CoserLewis. (2004). Masters of Sociological Thought, Rawat Publications, NewDelhi.

13. RaoShankar C.N. (2012).Sociology: principles with an Introduction toSocial Thought.S.Chand and Company Pvt.Ltd

BAHSECOVAC-103:INTRODUCTION TO ENVIRONMENTAL ECONOMICS

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours(3 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Course Objective: -This paper aims to make aware students about the importance of environment in economics and vice-versa. It helps them to know the ways of sustaining our natural resources by optimally allocating them for future use.

Course Outcomes: -This paper covers topics related to ecosystems, natural resources, biodiversity, conservation environmental pollution and policies.

Pedagogy of the Course Work:-80% Class room teaching.

20% Internal Assessment.

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Instructions for Paper-setter and candidates:

• The maximum marks for the paper will be 50. The question paper will be of 40 marks and continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.

• There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from eachunit. Each question will carry 10 marks (15x2=30).

UNIT – I

Introduction: Basic concepts of Environment, Economics and Ecology. Relation between Environment and Economy (inter linkages), the natural environment and the human economy. Envoirmental Externalities in Production and consumption. Managing externalities with taxes

and subsidies and Property Rights.Coase Theorem

UNIT -II

The openaccessandthepublicgoodproblem;commonpropertyresourcesandeconomicproblem.
Indian Law pertaining to Environmental protection .Environment Protection Act 1986.
Environmental Ethics: Issues and possible solutions, problems related to lifestyle, sustainable development ;consumerism and waste management .

Essential Readings:-

1. Maddala ,G.S. (2004). *Microeconomics; Theory and Practice*
2. Pindyck ,R &Rubinfeld, D : *Microeconomics* ,8th edition

Suggested Readings:

1. Eugine, T. (2005) .*Environmental Economics*. Varinda Publications, Delhi.
2. Goodstein, E.S. (2002).*Economics and the Environment* (third edition)John Wiley &Sons., Inc.
3. Hussen, Ahmed (2004) .*Principles of Environmental Economic* (second edition), Routledge, Taylore& Francis Group,London & New York.
4. Kolstad. C.D. (2011). *Intermediate Environmental Economics*

BAHSECOVAC-104 : INTRODUCTION TO DATA ANALYSIS USING EXCEL-II

Topology of the code	**Value Added Course**
Level of the Course	**100-199**
Course Duration /LTP (Lectures, Tutorials, Practical)	**L= 30 hours(2 hours a week for 15 weeks= 01 semester), T=0, P=0**
Credits	**2**
Max. Marks	**50 Marks = Theory 40 marks + Internal Assessment 10 marks**
Time	**2 hours**

Objective: The objective of the paper is to rigorously and comprehensively equip the students with applied data handling skills using spreadsheet based graphical and statistical tools of reasoning involved in empirical analysis of economic behaviour. The students would be evaluated at the end of each semester through subjective type questions/ answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

Instructions for Paper-setter and candidates:

• The maximum marks for the paper will be 50. The question paper will be of 40 marks and

continuous evaluation 10 marks. Time allowed will be 2 hours.

The paper-setter must put a note in the question paper in this regard.
• There shall be 5 questions in all.

The first question compulsory comprising 10 short answer type questions spread over the whole syllabus. The candidates are required to attempt 5 questions. Each question shall be of two marks (5x2= 10).

Rest of the paper shall contain two units. Two questions shall be asked from each unit and the candidates shall be given internal choice. The candidates shall attempt one question from eachunit. Each question will carry 10 marks (15x2=30).

Unit-I

Macro and Macro Recordings

Introduction to Excel Macros. Using Excel-Macro recording for data analyses (without using VB).

Unit-II

Mathematical and Statistical Analysis

Solution of matrices, Central Tendency and Dispersion measures, Correlation and regression analysis.

References:

Lloyd, Chris, J. (2011). Data Driven Business Decisions, John Wiley & Sons.

Levine, D.M., Stephen, D. and Szabat, K.A. (2016).Statistics for Managers Using Microsoft Excel, Pearson Education.

Day, Alastair L. (2013). Mastering Financial Mathematics in Microsoft Excel, Prentice Hall.
Piascik, Chester (2001). Applied Mathematics with Microsoft Excel, Pennsylvania State University.

Jelen, Bill (2013). Excel 2013: Charts and Grap Microsoft Excel. Que Publishing.

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STAT-M-151: INTRODUCTORYPROBABILITY

Programme Outcome: The aim of the course is to pay a special attention to applications of Probability Theory. The main focus of the course is to make the students familiar with various probability distributions and their uses in real life. The focus will be both on theoretical as well as practical approach.						
Course Outcome: On completion of this course students will be able to understand the concept of probability theory, random variable and their significance in real world. Students will get knowledge about variousdiscreteandcontinuousdistributionstosolveproblems.Studentswillunderstandtheusageof Central LimitTheorem.						
Class	**Sem.**	**CourseCode**	**Course Title**	**Credits= 6**		
B.A./B. Sc.	**II**	**STAT-M-151**	**INTRODUCTORYPROBABILITY**	**L=4**	**T=1**	**P=1**

Total Marks (Theory= 100 (Tutorial+ Practical= 50) Internal Assessment= 20%	
Total Hours= 105 hours (Theory=60 hours + Tutorial=15 hours +Practical= 30 hours)	
Time Allowed for Examination: Theory= 3 hours; Practical= 3 hours	
Instructions for Paper setter	There will be in all nine (9) questions, all of equal marks. The first question is **compulsory** and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit
Instructions for Candidate	The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each Unit**.**

Course Contents:

UNIT I

Probability: Introduction, random experiments, sample space, events independent, dependent, mutually exclusive, exhaustive, equally likely and algebra of events. Classical, statistical, and axiomatic approaches to probability, Conditional Probability, laws of addition and multiplication, theorem of total probability, Bayes' theorem and its applications,

Random Variables (for univariate distribution only): Discrete and continuous random variables, Probability massfunction(p.m.f.),Probabilitydensityfunction(p.d.f.),Cumulativedistributionfunction(c.d.f.)Illustrations ofrandomvariablesandtheir properties.Expectation,variance,momentsandmomentgeneratingfunction.

UNIT II

Standard probability distributions: Binomial, Poisson, geometric, negative binomial, hyper geometric, uniform, normal, exponential, beta, gamma and their applications. Fitting of Binomial, Poisson and Normal Distributions Chebyshev'sinequality ,Convergence in probability, Weak law of large numbers, Convergence in Distribution, De-Moivre Laplace and Lindeberg-Levy Central Limit Theorems (C.L.T.).

Notes:

1. **This course is not for the students of B.A./B.Sc. (General) Statistics Discipline Specific courses(Major/Core).AnyothercandidateshallofferthisminorsubjectinB.A./B.Sc.**
2. **Students are to be engaged to 7 hours of teaching per week (4 lectures for theory, one lecture for tutorial and 2 lectures forpractical).**
3. **Practical Examination will be held in the respective/samesemester.**

SUGGESTED READINGS:

1. Hogg, R.V., Tanis, E.A. and Rao J.M. (2009): Probability and Statistical Inference, Seventh Ed, Pearson Education,

NewDelhi.

2. Miller, Irwin and Miller, Marylees (2006): John E. Freund's Mathematical Statistics with Applications, (7th Edn.), Pearson Education,Asia.

3. Rohatgi, V. K. and Saleh A.K.M.E. (2010) : An Introduction to Probability Theory and MathematicalStatistics, WileyEastern.

4. Meyer, P.L. (1970): Introductory Probability and Statistical Applications, Oxford & IBH Publishing, New Delhi .

STAT-M-152: INTRODUCTORY PROBABILITY (PRACTICAL)

(Credit: 1)

Practical: Max. Marks: 25

Practical will be based on Paper STAT-M-152: INTRODUCTORY PROBABILITY.

Practical: Final Written Examination: 15 marks, Internal Assessment: 5 marks, viva-voce: 5 marks

Note for Practical: The practical question paper will consist of 5 questions. The students will be required to attempt any three questions in three *three*duration.

MAT-MI-161: Algebra

Credits: 3 (L=3; T=0; P=0)
Marks: 75 (Including Internal Assessment= 15)

Total hours: 45 (Theory=45) Total
Time Allowed for Examination: 3hrs.

Instructions for the Candidates and Paper Setters Candidates will be asked to attempt ve questions out of nine, carrying equal marks. Question No.1 spread

over the whole syllabus will be compulsory.

There will be two questions from each unit and the students will have to attempt one from each unit.

Course Outcome: The objectives are to introduce a system of linear equations and their solutions, polynomials and their roots and matrix theory. This course will acquaint students to algebra prior to the introduction of more abstract concepts like groups, rings and fields.

Unit-I

Review of system of linear equations, general theory of system of linear equations, rank of a matrix, row and column ranks of a matrix, system of homogenous and non-homogeneous linear equations.

Unit-II

Matrices, Rank of a matrix, Inverse of a Matrix, Elementary linear transformations, Determinants and their properties, Cramer's rule, Cayley-Hamilton Theorem.

Unit-III

A deeper look at Complex Numbers, taking roots of complex numbers, quick review of operations on polynomials, divisors and greatest common divisor, roots of polynomials, fundamental theorem, corollaries of fundamental theorem.

Unit-IV

Evaluating roots of polynomials of third and fourth degree, bounds of roots, Sturm's theorem, Descarte's rule of signs, approximation of roots.

(Scope as in Chapters 2, 3, 4, 5, and 9 of (A).)

Essential Textbook

(A) A. Kurosh Higher Algebra, MIR Moscow, 1982.

Further Readings

1. Shanti Narayan and P. K. Mittal, A textbook of Matrices S. Chand and Company Limited 2019.

MAT-MI-162: Differential Equations

Credits: 3

Max. Marks: 75 (Including Internal Assessment-15) Time allowed: 3hrs.

Candidates will be asked to attempt five questions out of nine, carrying equal marks. Question No.1 spread over the whole syllabus will be compulsory.
There will be two questions from each unit and the students will have to attempt one from each unit.
Objective: The main objective is to exhibit the techniques for obtaining solutions to ordinary differentialequations, the basic ideas and the theory behind those techniques. The emphasis will be to combine theabstract concepts with examples in order to intensify the understanding of the subject.

UNIT-I

Origin of Differential Equations, Basic definitions, Family of solutions, Geometric interpretation, Isoclines,Initial and boundary value problem. Basic Existence Theorem (Statement).

UNIT-II

Equations of Order One, Separation of variables, Exact Equations, Linear Equations, Integrating factors,Bernoulli's equation, Elementary applications.

UNIT-III

General linear equations: General solutions, Linear independence of solutions, differential operators. Linear equations with constant coefficients: Auxiliary equation.

UNIT-IV

Non-homogeneous equations: Method of undetermined coefficients, Variation of parameter method, Nonlinear Equations.

(Scope as in Ch. 1, sections 2.1-2.6, 5.1-5.5, 6.1-6.9, 7.1-7.3, 7.5, 8.3,8.5, 9.1-9.4 and Ch. 16 of (A)).
Essential Textbook

(A) Earl D. Rainville and P. E. Benediet, Elementary differential equations, Seventh Edition, Macmillian, Publishing Company, 1989.

Suggested Readings:-

1. S. L. Ross, Differential Equations, 3rd ed., John Wiley and Sons, India 2004.

2**.** W. E. Boyce and R. C. Diprima, Elementary differential equations and boundary value problems, Seventh Edition, John Wiley and Sons, Inc., 2001.

3. E. A. Coddington, An introduction to ordinary differential equations, Prentice- Hall Pearson, 2014.